

PARVEZ AHMED

Software Engineer · Backend & Microservices · Automation & Applied AI

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SUMMARY

Software Engineer with two years building production financial applications and microservices in a global custodian bank. Python, Java, Spring Boot, React, SQL. Modernised 16 Java services, cut an ISO 20022 data pipeline from nine minutes to under 50 seconds, and remediated 800+ CVEs. Builds measurable automation: LLM and RAG workflows over a Neo4j graph database, an ML matching model for reconciliation, and live open-source document-extraction and matching systems scored on held-out evaluation data.

PROFESSIONAL EXPERIENCE

Senior Analyst

Apr 2026 – Present

Northern Trust · Bengaluru, Karnataka · Promoted from Analyst, Software Engineer

- **Built an agentic RAG system over a legacy AS/400 codebase** using Azure OpenAI (GPT) models and a Neo4j knowledge graph, implementing LLM-driven Cypher query generation, graph retrieval, and inferencing through a custom chat interface that lets developers query decades-old business logic in natural language.
- **Trained a machine learning model to automate financial reconciliation** on historical manual-match data, learning to resolve one-to-many and many-to-many break patterns that the TLM auto-matching engine cannot, reducing manual reconciliation workload across operations teams.
- **Upgraded 16 Java microservices from Java 8 to Java 21 and Spring Boot 2.x to 3.x** with zero production downtime, and remediated 800+ CVEs by hardening dependencies and container configurations across enterprise banking infrastructure.

Analyst, Software Engineer

Aug 2024 – Apr 2026

Northern Trust · Bengaluru, Karnataka

- **Led zero-downtime migration of 21 Spring Boot microservices (v2.1.x → v2.7.x)**, resolving dependency conflicts across enterprise banking infrastructure.
- **Built a Python report-conversion tool** transforming large TLM HTML reports into optimised XLSX files, cutting load time from several minutes to under 5 seconds. Adopted company-wide by operations teams.
- **Engineered an ISO-to-CAMT.053 conversion pipeline** parsing large financial XML files into optimised chunks, reducing processing time from ~9 minutes to under 50 seconds (82% reduction).
- **Automated CI/CD pipelines via GitHub Actions**, cutting deployment time by 40% and improving release reliability across PCF-hosted services.
- **Handled RHEL7 → RHEL8 upgrade of Unix production servers** for a system with 30+ upstream dependencies.

Project Intern

Jun 2023 – Sep 2023

Northern Trust · Bengaluru, Karnataka

- **Improved frontend Lighthouse scores by 25%** through targeted audits, code-splitting, and accessibility fixes, and integrated REST APIs powering dynamic financial dashboards.

Project Collaborator

Oct 2022 – Mar 2023

Volkswagen Group New Academy · Chakan, Pune, Maharashtra

- **Built a full-stack Decision Support System (Django, React, PostgreSQL, Docker)** for sales and HR analytics across 100+ dealership branches, replacing manual CSV aggregation with on-demand real-time reporting.

PROJECTS

DocVal — Bank-Statement Extraction with Arithmetic Validation

Python · FastAPI · Docker

Live: docval-yy4s.onrender.com · Code: github.com/Sero01/docval

- **Built a Python service that converts PDF bank statements into typed JSON**, routing each page between native text-layer parsing and vision-LLM extraction, then validating the arithmetic so opening balance plus every transaction equals the closing balance.
- **Measured held-out F1 of 0.63 overall**, 0.92 on digital PDFs and 0.42 on dense scans, at roughly \$0.008 per document. 61 tests run behind a CI gate that fails the build on accuracy regression.

ReconMatch — Transaction Reconciliation & Break Classification

Python · matching engine

Live: reconmatch-aa9c.onrender.com · Code: github.com/Sero01/reconmatch

- **Built a deterministic matching engine** that aligns bank-statement lines to ledger entries through tiered rules (exact, date-windowed, split/partial, N:1 exact-sum batch), with a greedy assignment pass that prevents double-allocation and classifies every unmatched item as a typed break.
- **Resolved 88.65% of records on the BenchRec cash v1.0 held-out set** under strict exact-set scoring, against 62.45% for the published reference. Deterministic by design, because reconciliation output must be auditable. 51 tests.

Recommendation Engine & Platform Security — Gaming Media Startup

TypeScript · Next.js · Prisma

Live consumer web application · shared codebase, 12 contributors

- **Designed and shipped the discover recommendation engine:** 19 TypeScript modules across a three-stage retrieval, ranking and rerank pipeline, with an EMA taste model, a satisfaction controller and a session-scoped ring state machine. Verified against a 14-check offline simulation harness before a flag-gated rollout.
- **Found and fixed an unauthenticated account-takeover** in the password-reset route, then closed the same bug class across seven mutation routes by deriving identity from the session rather than the request body.

tiny-router — Can a 26M-Parameter Model Be a Safe First Stage?

Python · LLM evaluation

- **Tested whether a ~26M-parameter model can be the cheap first stage of an LLM workflow**, emitting one line of a closed typed action DSL that a hand-written validator and tier policy turn into execute, draft or escalate.
- **Built a 1,028-example evaluation corpus** with a risk/coverage sweep and ablation ladder. Calibration drove expected calibration error to near zero and changed unsafe executions by exactly zero; treating message bodies as untrusted cut them from 11 to 7.

Emma & Council — Local-First Agent Tooling

Python · FastAPI · Ollama

- **Emma** is a self-hosted agent on Gemma via Ollama with a plan-first PLAN, EXECUTE, SYNTH loop, six tools and a FastAPI server with SQLite persistence. **Council** runs a three-way terminal workflow across two coding agents and one codebase. Both run fully offline.

agentguides.dev — Weekly Publication on the AI Agent Ecosystem

Astro · Cloudflare Pages

- **Built the site and the cron-driven pipeline that publishes it every Sunday** without a missed week for six months: LLM leaderboards, model reviews and AI coding-tool comparisons, generated by automation I built and supervise.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, Bash, HTML, CSS

Backend & Web: Spring Boot, FastAPI, Django, Node.js, REST APIs, microservices, web applications, React, Next.js

Data & Databases: PostgreSQL, MySQL, SQLite, Neo4j graph database, Prisma, data modelling, XML/JSON/CSV processing, XLSX/Excel automation, data engineering, ISO 20022 / CAMT.053

AI / ML: LLMs, RAG, graph retrieval, agentic workflows, prompt engineering, ML model training, evaluation frameworks (held-out splits, calibration, risk/coverage), Azure OpenAI, Ollama

Cloud & DevOps: Docker, CI/CD with GitHub Actions, Git, Linux/RHEL, Azure, Pivotal Cloud Foundry, Cloudflare, automated testing (pytest, JUnit), performance optimisation, CVE remediation

Automation & Domain: Operational workflow automation, decision-support systems, enterprise system integration, financial reconciliation (TLM), custodian banking operations

EDUCATION

B.Tech, Computer Science Engineering

Sep 2020 – May 2024

Vishwanath Karad MIT World Peace University · Pune, Maharashtra · CGPA 9.55 / 10 · Top-6 finalist, Northern Trust Intern Hackathon

Class XII (CBSE) — Ryan International School, Raipur, Chhattisgarh · 95.6%

2020

CERTIFICATIONS

Claude Academy: Deploying Claude Enterprise with Confidence

Sep 2026

The five decisions that shape your rollout · Anthropic · Credential ID 4b526b6f15d738b986f0b98ae30323dc

Claude Code in Action

Sep 2026

Anthropic · Credential ID iadedge5dett